

Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

Revisiting KNN and Centroid-Based Classification: How Dataset Structure Influences Comparative Performance

By
Md. Hossain Akash

October 26, 2026

Masters Thesis

Submitted in partial fulfillment for the award of **Master of Science in Computing** to the Department of Computer Science & Applied Physics, Atlantic Technological University (ATU), Galway.

M.Sc. in Computing

Contents

1 Introduction **1**
- 1.1 The study we revisit . . . 1
- 1.2 Research questions . . . 2
- 1.3 Contributions . . . 2

2 Research Methodology **3**
- 2.1 Research approach . . . 3
- 2.2 Statistical inference . . . 4
- 2.3 Grading the evidence . . . 4
- 2.4 Software process and reproducibility . . . 4

3 Literature Review **5**
- 3.1 Nearest neighbour classification . . . 5
- 3.2 Centroid-based neighbourhoods . . . 5
- 3.3 The nearest centroid classifier . . . 6
- 3.4 Feature scaling and distance . . . 6
- 3.5 Feature weighting and metric learning . . . 6
- 3.6 Prototypes and the local–global spectrum . . . 7
- 3.7 How classifier comparisons mislead . . . 7
- 3.8 Data complexity and the gap . . . 8

4 Experimental Design **9**
- 4.1 Datasets . . . 9
- 4.2 Classifiers . . . 10
- 4.3 Evaluation protocol . . . 11
- 4.4 Structural measures . . . 12
- 4.5 Controlled experiments . . . 12

5 Results and Evaluation **14**
- 5.1 Replication on raw features . . . 14
- 5.2 The effect of standardisation . . . 14
- 5.3 Scale heterogeneity explains where the gap shrinks . . . 16
- 5.4 Controlled evidence for the mechanism . . . 17
- 5.5 A global reference: the nearest centroid classifier . . . 18
- 5.6 Skewness with separation held fixed . . . 19
- 5.7 Separability and the ceiling effect . . . 21
- 5.8 Structure-informed classifier selection . . . 21
- 5.9 Threats to validity . . . 22

6 Conclusion **24**

A Code, Data **30**

List of Figures

2.1 Five stages of the study 3

5.1 NCN − KNN gap on raw and standardised features 16
5.2 Gap shrinkage against scale heterogeneity 17
5.3 Synthetic scale sweep and Wine k sweep 18
5.4 Three-way comparison and gap against skewness 19
5.5 Skewness experiment with separation held fixed 20

List of Tables

4.1 Datasets as used in the experiments 10
4.2 Structural profile of the datasets 11

5.1 KNN and NCN accuracy: original, raw and standardised 15
5.2 Skewness experiment with separation held fixed 20
5.3 Separability sweep on symmetric Gaussian classes 21
5.4 Structure-informed selection guide 22

Abstract

A 2022 study compared the classical k-nearest neighbour rule (KNN) with its nearest centroid neighbourhood variant (NCN) on eighteen public datasets and reported a statistically significant advantage for NCN on two of them. The models in that study were trained on raw attributes, with no feature scaling. This thesis asks a simple question: does the advantage belong to the algorithm, or to the way the data were presented to it?

We replicated the comparison on fifteen of the original datasets using ten repetitions of stratified ten-fold cross-validation and a corrected resampled t-test. We then repeated every run with the features standardised inside each training fold. The raw-feature results reproduced the original pattern closely. After standardisation, the NCN advantage on Stars fell from 0.043 to 0.005, on Wine from 0.048 to 0.008 and on Cryotherapy from 0.079 to zero, while the advantage on Ionosphere (0.086) did not move. Across datasets, the size of this drop followed the heterogeneity of the raw feature scales (Spearman $\rho = 0.60$, $p = 0.019$). A synthetic experiment in which only the scale of one feature was changed reproduced the effect, and on Wine the raw-feature gap grew with k while the standardised gap stayed close to zero.

We then added the nearest centroid classifier (NC) as a global reference model. In a synthetic experiment that increased skewness while holding a centroid-based separability ratio near a fixed value, NC's shortfall from the Bayes-optimal accuracy grew to about twice that of KNN, while NCN stayed closer to the optimum than KNN at every level. We turn these findings into an evidence-graded guide for choosing among the three methods from a few structural measurements taken before training. The main lesson is plain: a classifier comparison is only as general as the preprocessing and the data it was run on.

Acknowledgments

I would like to thank my supervisor, Douglas Mota Dias, for clear and honest feedback throughout this project, and in particular for pushing me to test my claims more carefully than I first did.

I also thank the Department of Computer Science & Applied Physics at Atlantic Technological University, Galway, for the teaching and resources behind this work, and the maintainers of the UCI Machine Learning Repository for making the datasets available.

Finally, I thank my family and friends for their support during my studies.

Chapter 1

Introduction

The k-nearest neighbour rule (KNN) is one of the oldest ideas in machine learning and still one of the most used. It stores the training data, finds the k points closest to a query and lets them vote [1, 2]. There is almost nothing to train and the decision is easy to explain. The price is that everything depends on the distance. If the distance is misleading, the neighbours are the wrong neighbours. Many variants try to choose neighbours more carefully. One of them is the nearest centroid neighbourhood rule (NCN) [3, 4], which picks neighbours one at a time so that their centroid stays as close as possible to the query. A good neighbourhood, in this view, should *surround* the query rather than sit on one side of it.

1.1 The study we revisit

Mladenova and Valova [5] compared KNN and NCN with $k = 5$ on eighteen public datasets chosen to resemble small-business data: small samples, mixed attributes, outliers and imbalance. NCN was significantly better on two of them, Ionosphere and Stars. It is a clear study with a modest conclusion. What caught our attention was one sentence in the method: “All available attributes were used for each model training without any additional attribute processing.” For a method built on Euclidean distance, that sentence matters. When one attribute is measured in thousands and another in fractions, the first decides who the neighbours are. KNN and NCN use the distance differently, so a gap between them on raw data could come from the algorithms or from how each reacts to badly scaled features. The original design cannot tell these apart.

1.2 Research questions

RQ1 Does the NCN advantage reported on raw features survive when the features are standardised?

RQ2 If it changes, does the change follow the heterogeneity of the raw feature scales, and can it be reproduced under controlled conditions?

RQ3 Which measurable data properties can guide the choice between KNN, NCN and the simpler nearest centroid (NC) classifier, and how strong is the evidence for each?

1.3 Contributions

The thesis makes four contributions. It gives an independent replication of the KNN–NCN comparison on fifteen of the eighteen original datasets, using a test that accounts for overlapping training sets. It shows that most of the reported NCN advantage disappears under standard preprocessing, and that the size of the loss tracks feature-scale heterogeneity. The one advantage that survives, on Ionosphere, is on the dataset whose features were already on a common scale. It adds a skewness experiment, with a centroid-based separability ratio held fixed and each method measured against the Bayes-optimal accuracy, which shows why a global model such as NC is fragile where local models are not. Finally, it turns the evidence into a short, graded guide for choosing a classifier from the structure of the data.

We do not claim that the original study was wrong. Its numbers are reproducible, and we reproduce them. Our point is narrower. The conclusion holds for raw features, and a reader who standardises their data, as most practitioners do, should not expect the same result.

Chapter 2

Research Methodology

2.1 Research approach

This is a quantitative, experimental study built on *replication followed by intervention*. We first reproduce the original comparison as closely as we can. Then we change one thing, the preprocessing, and hold everything else fixed: datasets, folds, k and random seeds. If the result moves, the intervention moved it.

Real data show *that* something happens but rarely *why*, so each real-data claim is paired with a synthetic experiment that varies one property and holds the rest constant. The work ran in five stages, shown in Figure 2.1.

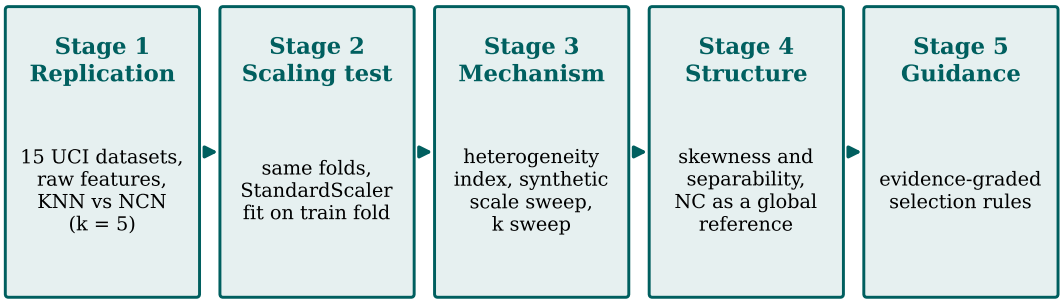


Figure 2.1: The five stages of the study. Each stage changes one factor relative to the previous one.

2.2 Statistical inference

Cross-validation folds share most of their training data, so a plain paired t-test on fold accuracies is too liberal [6]. We use the corrected resampled t-test of Nadeau and Bengio [7]. For J paired fold differences with mean $\bar{d}$ and variance s^2,

$$t = \bar{d} \Big/ \sqrt{\left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s^2}, \tag{2.1}$$

with $J - 1$ degrees of freedom. With 15 datasets, a Bonferroni correction gives a per-test threshold of $\alpha = 0.0033$. Across datasets we use the Wilcoxon signed-rank test [8] and Spearman's rank correlation, as recommended by Demšar [9], and a leave-one-out check where a result rests on one dataset.

2.3 Grading the evidence

Each recommendation carries a grade. **Strong (S)**: supported by a controlled experiment *and* by real data that survives correction, or true by construction. **Moderate (M)**: supported by a controlled experiment, with only a trend in real data. **Weak (W)**: rests on one dataset or a small effect, and is treated as a hypothesis.

2.4 Software process and reproducibility

The code uses Python, scikit-learn [10] and SciPy [11], and was developed iteratively, one dataset first and then all. NCN was implemented from its definition [4] and checked to match KNN exactly at $k = 1$. Seeds are fixed. Six experiment scripts, sharing one small module for loading, evaluation and measurement, reproduce every number in this thesis, and a seventh builds every figure from their output. Following current advice on reproducible machine-learning research [12, 13], the code, data, a README and this thesis are public at `https://github.com/mdakash41/thesis-comparative-analysis-knn-nc` [14]. Appendix A describes the repository.

Chapter 3

Literature Review

3.1 Nearest neighbour classification

The nearest neighbour rule goes back to Fix and Hodges [1]. Cover and Hart [2] showed that, as the sample grows, the error of the one-nearest-neighbour rule is at most twice the Bayes error. In practice samples are small, and performance depends on two choices: the value of k and the distance used to find neighbours [15]. Both interact with the data. In high dimensions, distances between points become similar and the idea of a "nearest" neighbour weakens [16]. Something similar, in a milder form, happens when a few features dominate the distance simply because of their units.

3.2 Centroid-based neighbourhoods

Chaudhuri [3] argued that a neighbourhood should be judged not only by closeness but by how its points are arranged around the query: the neighbours should be close, and their centroid should be as close to the query as possible. Sánchez, Pla and Ferri [4] turned this into the k-NCN classifier and reported gains over KNN on several benchmarks, later refining it with heuristic modifications [17]. Related rules combine the centroid idea with local class means [18, 19], with gains reported mostly on small samples with outliers. These papers share an assumption that is rarely stated: the features are already on comparable scales. The benefit of NCN is explained geometrically, as a better-balanced neighbourhood, and that explanation only makes sense if the geometry itself is sensible.

3.3 The nearest centroid classifier

The nearest centroid classifier replaces each class by its mean and assigns a query to the closest one. Prediction costs $O(Cd)$ for C classes in d dimensions, compared with $O(nd)$ for KNN. A shrunken version works very well on high-dimensional gene-expression data [20]. Its weakness is the assumption that one mean describes a class. This holds for compact, roughly symmetric classes and fails for skewed or multimodal ones, where the mean sits away from most of the points [15]. NC was not part of the original study. We include it because it sits at the global end of the local-versus-global spectrum, which lets us ask which data properties hurt global summaries and which do not.

3.4 Feature scaling and distance

Distance-based methods are known to be sensitive to scale. Singh and Singh [21] compared fourteen normalisation methods and found that the choice changes classifier accuracy, with KNN among the most affected, and Ahsan et al. [22] reached a similar conclusion on medical data. What has received less attention is whether scaling affects two closely related distance-based methods *differently*. If it does, comparing them on unscaled data measures their sensitivity to scale as much as their ability to classify. Scaling must also be fitted on the training fold only. Fitting it on the whole dataset leaks test information, which is one of the most common causes of irreproducible results in applied machine learning [23].

3.5 Feature weighting and metric learning

Standardisation is the simplest answer to unequal scales: it gives every feature the same spread, whether or not the feature is useful. A large body of work goes further and weights features by how much they help classification. Wettschereck, Aha and Mohri [24] reviewed feature weighting for lazy learners such as KNN and found that weights learned from the class labels tend to help, most clearly when some features are irrelevant. Metric learning takes the same idea to a full distance: large-margin nearest neighbour learning [25] learns a linear transformation in which the k nearest neighbours of each point share its class. Weighting can also act on the vote rather than the features, as in the distance-weighted KNN rule of Dudani [26], where closer neighbours count for more.

This literature matters here for two reasons. First, it shows that the choice of distance is part of the model, so two classifiers compared under an arbitrary distance are not compared on equal terms. Second, it points at a limit of our own heterogeneity index (Section 4.4), which looks only at the spread of each feature

and not at its relevance. A supervised weighting would treat the Breast Cancer case differently, and we return to this in Chapter 6.

3.6 Prototypes and the local–global spectrum

KNN, NCN and NC can be placed on one line. KNN keeps every training point and decides locally. NC keeps one prototype per class and decides globally. Prototype selection and generation methods fill the space between them, keeping a subset or a summary of the training set [27]. García et al. compared dozens of such methods and found a clear trade-off between how much of the training set is kept and how much accuracy is retained. NC is the most aggressive reduction of all, which is why it is cheap and why it is fragile when one mean per class is a poor summary. Seen this way, our question about NC is a question about when the most aggressive reduction is safe.

3.7 How classifier comparisons mislead

Hand [28] argued that many reported improvements in classifiers are smaller than they look, because they depend on details of the data and evaluation that do not transfer. The no-free-lunch results [29] and large benchmark studies [30] make the same point from different directions. On the statistical side, the paired t-test on cross-validation folds has an inflated Type I error [6]. Nadeau and Bengio [7] corrected it, and Demšar [9] recommended non-parametric tests across datasets. Benavoli et al. [31] went further and argued for Bayesian comparisons, which report how probable a practically meaningful difference is rather than whether a null hypothesis can be rejected. This applies directly to the original study, which used an uncorrected t-test on ten folds for each of eighteen datasets. At $\alpha = 0.05$, the chance of two or more significant results among eighteen independent tests when there is no real difference is about 0.23. On their own, two significant results are weak evidence.

A related concern is reproducibility. Gundersen and Kjensmo [13] surveyed 400 papers from leading AI conferences and found that few documented their method, data and experiments well enough to be reproduced. The original study is a mild case: it does state that no attribute processing was applied, which is exactly what made this thesis possible.

3.8 Data complexity and the gap

If no classifier wins everywhere, the next question is whether the winner can be predicted from the data. Ho and Basu [32] proposed complexity measures such as feature overlap and boundary length, and Lorena et al. [33] review the field. Most of these measures describe the class boundary. Fewer describe simple distributional properties, such as skewness or scale heterogeneity, that can be computed before training and that relate directly to how distance-based methods work. What the literature does not establish is whether the reported advantage of NCN survives the preprocessing most practitioners apply, and which simple data properties decide when a local or a global method should be preferred. This thesis addresses that gap.

Chapter 4

Experimental Design

4.1 Datasets

We used fifteen of the eighteen datasets in the original study. Three were not available to us in a form we could match: SRBCT, Beavers and Algerian Forest Fires. Breast Cancer was taken from the Wisconsin Diagnostic version distributed with scikit-learn. All others came from the UCI repository [34]. Rows with missing values were removed, non-numeric attributes were dropped and class labels were integer-encoded. Table 4.1 gives each dataset as it entered the experiments, next to the sizes in Table I of the original paper.

Five of our datasets differ from the original's, and the differences matter for the replication.

- **Thyroid** is the UCI thyroid ANN data (`ann-train` and `ann-test` combined: 7,200 records, 21 attributes, 3 classes). The original used new-thyroid (215 records, 5 attributes, 3 classes). These are different datasets, so our Thyroid results do not replicate the original's Thyroid row.
- **Absenteeism** uses absenteeism time in hours as the target, which takes 19 distinct values. The original reports 28 classes, which matches the reason-for-absence attribute instead, so our Absenteeism is a different task. Our copy also keeps the employee ID column as an attribute, in line with "all available attributes".
- **Lenses** has 3 of the 4 attributes in our copy.
- **Hepatitis** and **Lung cancer** hold only the complete records, 80 of 155 and 27 of 32. Incomplete records were removed before our experiments began.

Ecoli matches in size; we count 7 numeric attributes against the original's 8, presumably because the eighth is the non-numeric sequence name. Its accuracy does

Table 4.1: Datasets as used in the experiments, after cleaning: n records, d numeric attributes, C classes, the number of cross-validation folds and the paired accuracy values this gives over ten repetitions. The last three columns are from Table I of the original paper. A dagger marks a dataset that differs from the original's.

	As used here					Original		
Dataset	n	d	C	Folds	Pairs	n	d	C
Ecoli	336	7	8	2	20	335	8	8
Glass	214	9	6	9	90	214	9	6
Hepatitis†	80	19	2	10	100	155	19	2
Ionosphere	351	34	2	10	100	351	34	2
Iris	150	4	3	10	100	150	4	3
Lenses†	24	3	3	4	40	24	4	3
Lung cancer†	27	56	3	8	80	32	56	3
Soybean	47	35	4	10	100	47	35	4
Wine	178	13	3	10	100	178	13	3
Absenteeism†	740	20	19	2	20	740	19	28
Thyroid†	7200	21	3	10	100	215	5	3
Cryotherapy	90	6	2	10	100	90	6	2
Stars	240	4	6	10	100	240	4	6
Pima diabetes	768	8	2	10	100	768	8	2
Breast cancer	569	30	2	10	100	569	30	2

not match (Section 5.1). Table 4.2 gives the structural profile of each dataset, using the measures defined in Section 4.4.

4.2 Classifiers

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ be the training set and $\mathbf{q}$ a query. All three methods use Euclidean distance.

KNN. Find the k training points closest to $\mathbf{q}$ and predict the majority label among them. We use $k = 5$, as in the original study.

NCN. Build the neighbourhood greedily [4]. The first neighbour is the nearest point to $\mathbf{q}$. At step j, with neighbours $\mathbf{n}_1, \ldots, \mathbf{n}_{j-1}$ already chosen, the next neighbour is

$$\mathbf{n}_j = \arg \min_{\mathbf{x} \in \mathcal{D} \setminus \{\mathbf{n}_1, \ldots, \mathbf{n}_{j-1}\}} \left\| \mathbf{q} - \frac{1}{j} \Big(\mathbf{x} + \sum_{m=1}^{j-1} \mathbf{n}_m \Big) \right\|. \tag{4.1}$$

Table 4.2: Structural profile of the datasets. Het. is the scale-heterogeneity index on raw features; skewness and kurtosis are mean absolute values on raw features; separability, compactness and modality are computed on standardised features.

Dataset	Het.	|Skew|	|Kurt|	Separability	Compactness	Modality
Ecoli	0.90	0.91	4.41	3.62	0.62	2.25
Glass	2.68	1.48	4.33	2.48	0.81	4.17
Hepatitis	2.33	1.18	2.19	3.15	0.90	4.00
Ionosphere	0.32	0.48	0.60	2.72	0.97	4.50
Iris	0.61	0.30	0.52	3.42	0.46	1.00
Lenses	0.00	0.67	1.34	2.10	0.52	2.67
Lung cancer	0.56	0.88	1.44	4.97	0.81	1.00
Soybean	0.82	0.74	2.07	7.70	0.24	1.00
Wine	3.40	0.63	0.98	4.37	0.70	5.00
Absenteeism	2.47	0.75	2.01	3.48	0.66	2.47
Thyroid	1.79	6.26	167.85	3.13	0.83	5.67
Cryotherapy	2.42	0.92	1.90	2.12	0.88	5.00
Stars	4.23	1.12	3.41	2.63	0.33	2.83
Pima diabetes	2.54	1.06	3.12	1.43	0.99	5.50
Breast cancer	5.33	1.34	5.18	6.63	0.84	2.00

The prediction is the majority label among $\mathbf{n}_1, \ldots, \mathbf{n}_k$. The first step is the same as in KNN, so NCN and KNN give identical predictions when $k = 1$. This is a useful check on any implementation.

NC. Compute one centroid $\boldsymbol{\mu}_c$ per class and predict $\hat{y} = \arg\min_c \|\mathbf{q} - \boldsymbol{\mu}_c\|$.

4.3 Evaluation protocol

Each dataset was evaluated with ten repetitions of stratified ten-fold cross-validation. Stratification needs every class in every fold, so when the smallest class had fewer than ten members the number of folds was reduced to the size of that class, with a minimum of two. This happened on five datasets (Table 4.1): Glass (9 folds), Lung cancer (8), Lenses (4), Ecoli (2) and Absenteeism (2). Ten repetitions therefore give 100 paired accuracy values per method on ten datasets, and 90, 80, 40, 20 and 20 on those five. On Absenteeism one class has a single record, so it cannot appear in both folds, and with two folds every model is trained on half the data. Results on Ecoli and Absenteeism are the least precise in the study. The corrected test in Equation (2.1) uses the actual training and test sizes, so its p-values account for the reduced fold counts. KNN and NCN were

always evaluated on the same folds. The experiment was run twice. In the *raw* condition, the features were used as given, which matches the original protocol. In the *standardised* condition, a `StandardScaler` was fitted on each training fold and applied to both the training and the test fold. Nothing else changed between the two runs. On real data NCN was computed exactly. In the synthetic experiments, to keep run time manageable, the NCN search was restricted to the 60 nearest candidates of each query. This is an approximation, but at $k = 5$ the greedy choice in Equation (4.1) seldom reaches that far.

4.4 Structural measures

Scale heterogeneity. For raw features with standard deviations $\sigma_1, \ldots, \sigma_d$ (constant features excluded),

$$H = \log_{10} \frac{\max_f \sigma_f}{\min_f \sigma_f}. \tag{4.2}$$

$H = 0$ means all features vary on the same scale, and $H = 3$ means the most variable feature has a spread one thousand times larger than the least variable one. The index is deliberately crude. It ignores which features carry class information, which turns out to matter (Section 5.3).

Shape. Skewness and excess kurtosis are computed per feature and per class on raw features, and their absolute values are averaged. Scaling does not change shape, but raw values avoid any doubt.

Separability and compactness. On standardised features, separability is the mean distance between pairs of class centroids divided by the mean within-class standard deviation. Compactness is the mean within-class standard deviation itself, so a lower value means tighter classes.

Modality. For each class we fitted Gaussian mixtures with one to six components and chose the number by the Bayesian information criterion [35]. The average over classes is reported.

4.5 Controlled experiments

Scale sweep. Two Gaussian classes of 150 points with five features. Four features carry a mean shift of 0.9 and the fifth only 0.3. The weak fifth feature is multiplied by a factor s of 1, 2, 5, 10, 20, 50, 100 or 300, and nothing else changes. Both conditions use five repetitions of ten-fold cross-validation.

k **sweep.** On Wine, which had the clearest raw gap that later vanished ($p = 0.042$), both conditions were repeated for $k \in \{1, 3, 5, 9, 15\}$.

Skewness with fixed separation. An earlier version of this experiment increased skewness by widening a log-normal distribution, which also reduced class separation, so the two effects were confounded. We redesigned it. Each feature of class 0 is drawn from a log-normal distribution with shape $\sigma \in \{0.1, 0.3, 0.5, 0.8, 1.2, 1.5\}$ and standardised with its *theoretical* mean and variance. Class 1 is the same distribution shifted by δ on every feature, with δ set by bisection on 20,000 points per class so that the measured separability equals 2.2 at every level. What is held fixed is this centroid-based ratio, not the difficulty of the problem: a skewed class has a sharp edge, which makes the classes easier to tell apart at the same ratio. Because both class densities are known exactly, we also estimated the Bayes-optimal accuracy at each level on 200,000 points per class, and we report each method's shortfall from it. We used four features, 300 points per class, five datasets per level and two repetitions of ten-fold cross-validation, and measured separability and compactness on every dataset as a check.

Separability sweep. Two symmetric Gaussian classes in four dimensions, with the distance between centres varied from 0.5 to 4.0.

Chapter 5

Results and Evaluation

5.1 Replication on raw features

Table 5.1 sets the original accuracies next to ours. On the nine datasets that match the original in size and attributes (Table 4.1, no dagger, Ecoli aside), our raw-feature accuracies differ from the published ones by 0.012 for KNN and 0.016 for NCN on average. On the eight of them where the original reported a gap, ours points the same way on seven. The match is closest on the datasets that carry the original conclusion: Ionosphere (0.844/0.931 against 0.85/0.93), Stars (0.636/0.680 against 0.61/0.70), Wine (0.696/0.744 against 0.72/0.76) and Cryotherapy (0.696/0.774 against 0.67/0.79). Ecoli is the exception: it matches in size, but our KNN accuracy of 0.85 is far from the reported 0.42. We suspect a different version or encoding of the data, but we could not confirm this. Rows marked with a dagger come from datasets that differ from the original's (Section 4.1), so they are not like-for-like comparisons.

One discrepancy sits in the original paper itself. Its text gives the Cryotherapy difference as 8%, while its Figure 4 is labelled 0.67 and 0.79, a difference of 12 points. We took our values from the figure. Our own raw gap, 0.079, happens to agree with the text.

Under the corrected test, Stars ($p = 0.051$), Cryotherapy ($p = 0.052$) and Wine ($p = 0.042$) sit near the 0.05 line, and none survive Bonferroni correction. Only Ionosphere and Thyroid do.

5.2 The effect of standardisation

Figure 5.1 shows what happens after standardisation. On Cryotherapy, Wine and Stars, the three datasets with the largest raw gaps after Ionosphere, the gap collapses to 0.000, 0.008 and 0.005. KNN is the method that gains: on Stars it

Table 5.1: KNN and NCN accuracy ($k = 5$). "Original" values are read from the published figure [5]. Our results use ten repetitions of stratified cross-validation (folds as in Table 4.1). A dagger marks a dataset that differs from the original's. Δ = NCN − KNN. An asterisk marks significance under the corrected resampled t-test after Bonferroni correction ($p < 0.0033$).

	Original		Raw features (ours)			Standardised (ours)		
Dataset	KNN	NCN	KNN	NCN	Δ	KNN	NCN	Δ
Ionosphere	.85	.93	.844	.931	+.086*	.850	.937	+.086*
Cryotherapy	.67	.79	.696	.774	+.079	.896	.896	.000
Wine	.72	.76	.696	.744	+.048	.967	.975	+.008
Stars	.61	.70	.636	.680	+.043	.986	.991	+.005
Glass	.66	.73	.662	.693	+.031	.648	.686	+.038
Pima diabetes	.71	.73	.718	.740	+.023	.738	.738	−.000
Soybean	.98	.96	.979	.994	+.015	1.00	1.00	.000
Absenteeism†	.34	.32	.357	.372	+.015	.378	.400	+.022
Thyroid†	.93	.96	.940	.947	+.007*	.950	.960	+.010*
Lenses†	.73	.73	.821	.825	+.004	.767	.808	+.042
Breast cancer	.93	.93	.934	.937	+.003	.968	.964	−.005
Ecoli	.42	.42	.847	.848	+.001	.846	.844	−.003
Iris	.97	.96	.963	.960	−.003	.948	.959	+.011
Hepatitis†	.72	.76	.803	.793	−.010	.865	.864	−.001
Lung cancer†	.58	.55	.625	.518	−.107	.569	.499	−.070

rises from 0.636 to 0.986 and on Wine from 0.696 to 0.967. NCN improves too, but by less, because it was already partly protected. The raw-feature "advantage" of NCN was, to a large extent, a smaller penalty for bad scaling.

Two results do not follow this pattern. The Ionosphere gap does not move at all: +0.0863 raw and +0.0864 standardised, significant in both. Ionosphere's features are already on a common scale in $[-1, 1]$ ($H = 0.32$), so standardising them changes almost nothing. This is what we would expect if the advantage reflects class structure rather than units, and we regard it as genuine. Thyroid is also significant in both conditions, but at about one percentage point the effect is small in practical terms, and our Thyroid is not the original's dataset.

The largest gap in the table goes against NCN: Lung cancer, −0.107 raw and −0.070 standardised. We read it as noise. Lung cancer has 27 complete records and 56 attributes, so each of its eight folds tests three or four records, and one misclassification moves a fold's accuracy by 25 to 33 points. Neither gap is close to significant ($p = 0.21$ and $p = 0.49$).

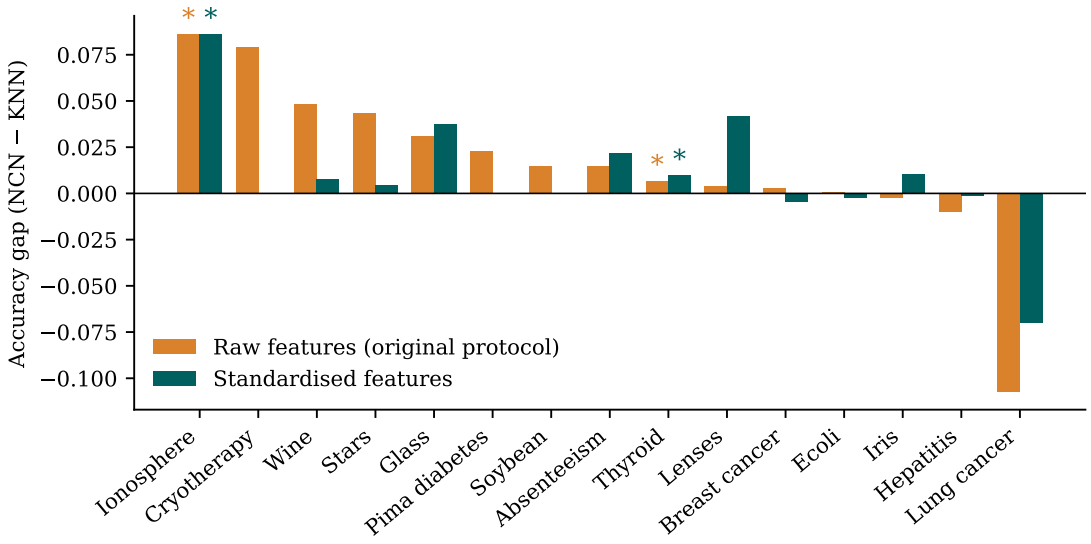


Figure 5.1: NCN − KNN accuracy gap on raw and standardised features, on the same folds. Asterisks mark gaps significant after Bonferroni correction.

Across all fifteen datasets, the mean absolute gap falls from 0.032 on raw features to 0.020 after standardisation. A Wilcoxon signed-rank test on the paired gaps is not significant ($W = 50$, $p = 0.60$), and neither is the same test on absolute gaps ($p = 0.28$). Standardisation does not shrink the gap everywhere; on Lenses and Glass it grows slightly. The effect is concentrated in particular datasets, and the next section asks which.

5.3 Scale heterogeneity explains where the gap shrinks

If bad scaling produces the raw gap, the gap should shrink most where the raw scales differ most. Figure 5.2 plots the shrinkage (raw gap minus standardised gap) against the heterogeneity index H. The rank correlation is positive and significant (Spearman $\rho = 0.60$, $p = 0.019$). The linear correlation is similar in size but just misses the conventional threshold (Pearson $r = 0.51$, $p = 0.054$). The datasets with the largest shrinkage, namely Cryotherapy, Wine and Stars, all have H above 2.4.

The relationship is not perfect, and the exceptions are informative. Breast Cancer has the highest heterogeneity ($H = 5.33$) but almost no raw gap to lose. Glass, Hepatitis and Absenteeism also have H above 2 without shrinkage. Our reading is that heterogeneity is close to a necessary condition for a scaling artefact,

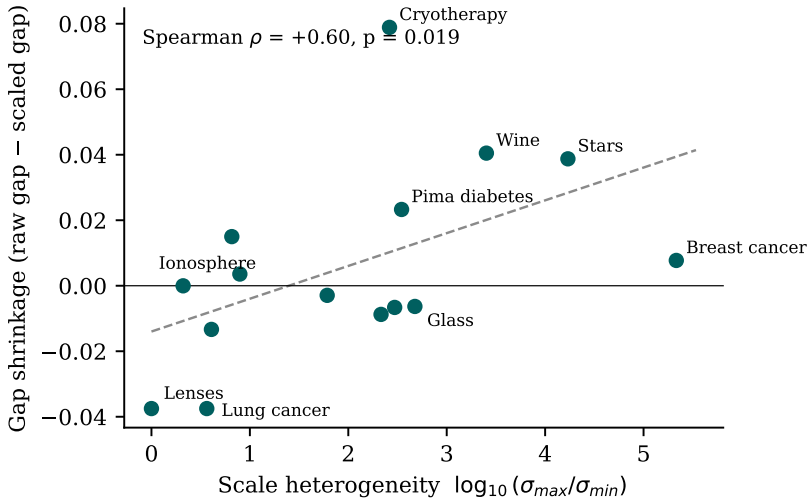


Figure 5.2: Gap shrinkage against the scale-heterogeneity index of the raw features. The dashed line is a least-squares fit for reference only.

but not a sufficient one. What matters is whether the features that dominate the distance are also the ones that separate the classes. In Breast Cancer the large-scale features, such as cell area, are among the most informative, so letting them dominate the distance costs little. The index in Equation (4.2) cannot see this, which is its main limitation.

5.4 Controlled evidence for the mechanism

A correlation on fifteen datasets does not establish a cause. In Figure 5.3(a), only the scale of one weak synthetic feature changes. On standardised data the gap is flat at -0.009, because standardising undoes the manipulation. On raw data the gap is near zero up to a factor of 10, then climbs to $+0.081$ at 50 and $+0.118$ at 100. This is the pattern seen in Wine and Stars, now produced on purpose. At a factor of 300 the gap falls back to $+0.027$. By then one feature dominates the distance so completely that both methods are, in effect, classifying on that feature alone, and there is little room left for any neighbourhood rule to differ.

Panel (b) shows the same effect on real data from another angle. At $k = 1$ the gap is exactly zero, as it must be. On raw Wine features it then grows steadily with k, reaching $+0.103$ at $k = 15$. On standardised features it stays within ± 0.012 at every k. A plausible explanation is this. When one feature dominates the distance, KNN's nearest points lie along that one axis, and as k grows the neighbourhood stretches along it into other classes. NCN's centroid

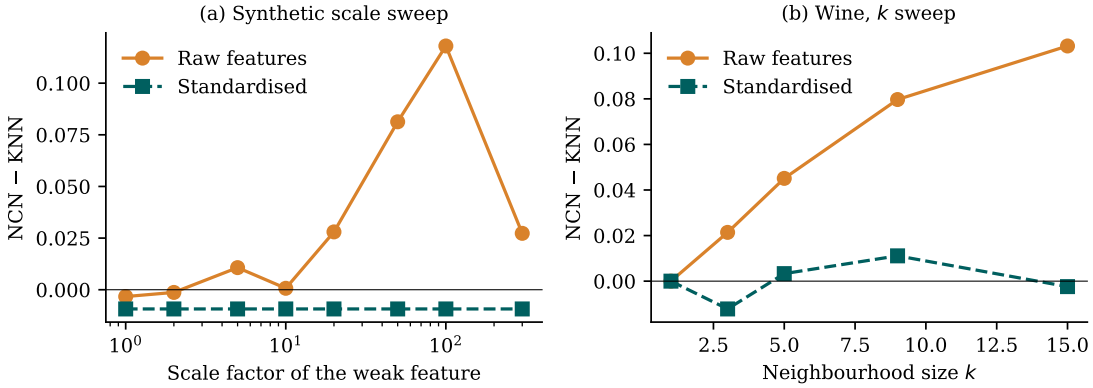


Figure 5.3: (a) Synthetic data: NCN − KNN gap as the scale of one weak feature grows. (b) Wine: gap as a function of k. In both, the gap exists only on raw features.

criterion pulls in points from both sides of the query and resists this stretching. Once the features are standardised, there is no dominant axis, and the two rules choose similar neighbours. This is an interpretation that fits both panels, not a proven mechanism.

So, for RQ1, most of the NCN advantage does not survive standardisation, with Ionosphere the clear exception. For RQ2, the loss tracks scale heterogeneity, and manipulating scale alone reproduces it.

5.5 A global reference: the nearest centroid classifier

We now move from preprocessing to structure. Figure 5.4(a) compares KNN, NCN and NC on standardised features. NCN beats KNN on 8 of 15 datasets and ties on two, but the mean gain is only 0.009. NC beats KNN on just 3 of 15 (Lenses, Lung cancer and Wine) and loses 0.053 on average. Its worst results are on Thyroid (-0.238), Absenteeism (-0.168) and Glass (-0.161).

Panel (b) suggests skewness is part of the story: across the fifteen datasets the NC gap correlates negatively with skewness ($r = -0.58$, $p = 0.023$) and the NCN gap does not ($r = -0.05$). We should not lean on this. The rank correlation is weaker ($\rho = -0.43$, $p = 0.108$), the result does not survive a Bonferroni correction over the five properties tested ($\alpha = 0.01$), and removing Thyroid alone drops it to $r = -0.25$ ($p = 0.40$). On real data, this is a trend and no more. That is why the controlled experiment matters.

Skewness is also not the whole story. NC's second- and third-worst results, Absenteeism (-0.168) and Glass (-0.161), have moderate skewness (0.75 and

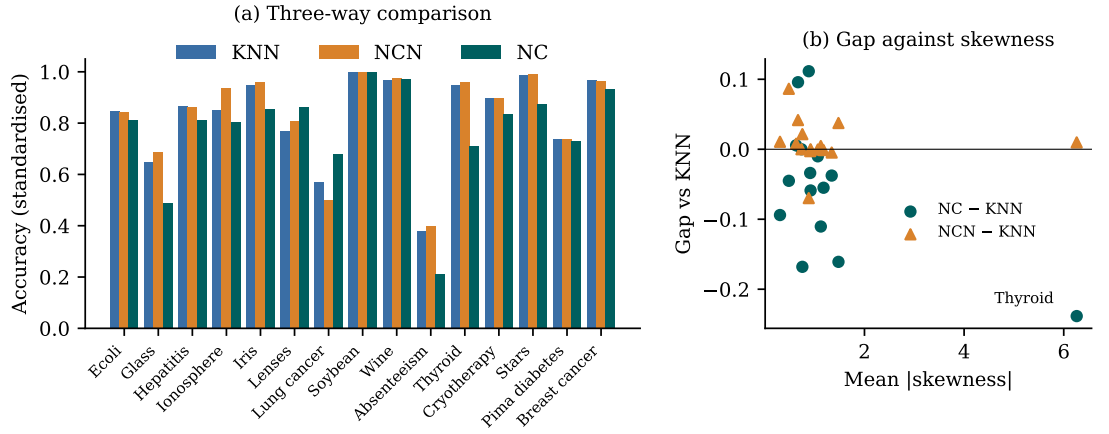


Figure 5.4: (a) Accuracy of KNN, NCN and NC on standardised features. (b) Gap to KNN against mean absolute skewness. Thyroid, the most skewed dataset, is also NC's worst case.

1.48), and a skewness rule would not flag Absenteeism at all. Both have structure that one mean per class handles badly. Absenteeism has 19 classes, eight of them with three records or fewer, so those centroids rest on almost no data. Glass has six classes, two of them with 9 and 13 records, and its classes look multimodal (4.17 mixture components per class on average), so one mean cannot describe a class made of several clusters.

5.6 Skewness with separation held fixed

Table 5.2 and Figure 5.5 report the redesigned experiment. Up to $\sigma = 0.8$, the measured separability stays between 2.15 and 2.27 and compactness between 0.87 and 0.88. At the two highest levels it drifts up to 2.54 and 2.63, because heavy tails make the sample standard deviation underestimate the true one. The control therefore holds up to $\sigma = 0.8$, and only approximately beyond.

Holding the ratio fixed did not hold the difficulty fixed. The Bayes-optimal accuracy rises from 0.868 at $\sigma = 0.1$ to 0.995 at $\sigma = 0.8$ and 1.000 beyond. A strongly skewed class has a sharp edge, which makes it easier to separate at the same ratio. This is why all three accuracies rise, and why we judge each method by its shortfall from the Bayes rate rather than by its accuracy.

On that scale the result is clear. At $\sigma = 0.1$ NC matches the Bayes rate (a shortfall of -0.001, within noise), as expected for near-symmetric classes whose best boundary is almost linear. From $\sigma = 0.5$ on, NC's shortfall is 0.081, 0.089, 0.038 and 0.034: about twice KNN's (0.047, 0.043, 0.019, 0.011), and three times

at $\sigma = 1.5$. NCN's shortfall is smaller than KNN's at every level. Measured against KNN instead, NC falls from $+0.014$ to -0.046 by $\sigma = 0.8$, while NCN stays level or slightly ahead, by up to $+0.019$.

Table 5.2: Skewness experiment with the centroid separability ratio held near 2.2 (mean over 5 datasets × 20 folds per level). Bayes = accuracy of the Bayes-optimal rule, estimated on 200,000 points per class; shortfall = Bayes − accuracy.

					Accuracy			Shortfall from Bayes		
σ	|Skew|	Sep.	Comp.	Bayes	KNN	NCN	NC	KNN	NCN	NC
0.1	0.34	2.27	0.87	.868	.856	.858	.870	.013	.010	−.001
0.3	0.91	2.19	0.88	.903	.867	.871	.863	.036	.032	.040
0.5	1.66	2.15	0.88	.953	.906	.916	.873	.047	.038	.081
0.8	3.09	2.21	0.87	.995	.953	.971	.907	.043	.024	.089
1.2	5.33	2.54	0.84	1.000	.981	.987	.962	.019	.013	.038
1.5	7.23	2.63	0.81	1.000	.989	.992	.966	.011	.008	.034

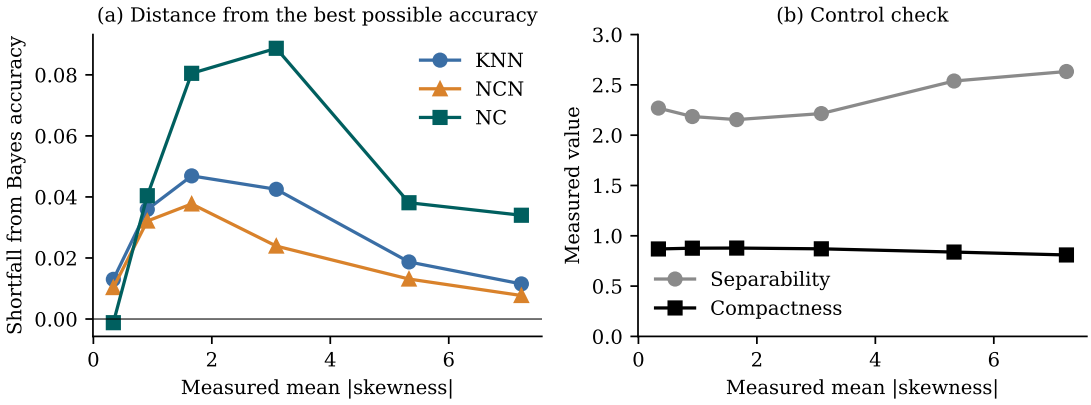


Figure 5.5: (a) Shortfall from the Bayes-optimal accuracy as skewness increases with the separability ratio held near 2.2. (b) Measured separability and compactness at each level.

The reason NC falls further from the optimum is simple. In a right-skewed class most points sit to the left of the mean, and a long tail drags the mean to the right. NC places its boundary halfway between two means that no longer sit where the data are. KNN and NCN never compute a mean over the whole class, so they follow the dense region. NCN goes one step further. By balancing its neighbours around the query, it copes slightly better with the asymmetric density near the boundary.

5.7 Separability and the ceiling effect

On symmetric Gaussian classes (Table 5.3) the optimal boundary is linear, so NC leads KNN at every level, but its lead shrinks from +0.058 to +0.007 as all methods approach perfect accuracy. Soybean (separability 7.70, all three score 1.00) and Wine (4.37, within 0.008) fit this picture.

The rest of the real data does not support a simple separability threshold. Of the nine datasets with separability above 3, NC loses to KNN on six, by up to 0.238 (Thyroid). Two things explain most of this. Our measure averages the distance over all pairs of classes, so one distant class can hide two close ones: in Iris, setosa sits far from the other two species, which overlap, and NC trails KNN by 0.094. And several of these datasets are skewed (Thyroid), have many small classes (Absenteeism) or have classes that look multimodal. High separability helps NC only when the classes are also symmetric and every pair of them is well apart.

Table 5.3: Separability sweep on symmetric Gaussian classes.

Centre distance	Separability	KNN	NCN	NC	NCN−KNN	NC−KNN
0.5	0.57	.550	.550	.607	+.001	+.058
1.0	1.06	.638	.652	.696	+.013	+.057
1.5	1.56	.734	.748	.778	+.014	+.045
2.0	2.06	.823	.821	.848	−.002	+.026
3.0	3.06	.927	.925	.938	−.002	+.011
4.0	4.06	.975	.979	.982	+.004	+.007

What we set aside. Kurtosis gave a clear null (KNN and NC above 0.97, gap within ±0.004). The compactness effect appeared in only one configuration. The modality design placed classes further apart than their modes, so it could not detect an effect. None of the three enters the guide.

5.8 Structure-informed classifier selection

Table 5.4 turns the findings into a graded guide. It shortens the search; it does not replace validation. Each row names the real datasets that support it and, where our own data contain them, the cases it gets wrong.

In practice: compute H on the raw training features and standardise within folds if it exceeds about 2; compute skewness, the number and size of the classes, and separability, preferably for the closest pair of classes rather than the average;

Table 5.4: Structure-informed selection guide. Grades: S = strong, M = moderate, W = weak (Section 2.3).

Data condition	Recommendation	Evidence	Grade
Raw scales differ widely ($H > 2$)	Standardise within folds before comparing; ignore raw-feature gaps.	Scale sweep; $\rho = 0.60$, $p = 0.019$; Wine k sweep	S
$k = 1$	NCN and KNN are identical.	By construction; Wine gap = 0	S
Mean |skewness| above about 1.5	Avoid NC; use KNN or NCN.	Skewness sweep (NC's shortfall from Bayes about twice KNN's); real-data trend rests on Thyroid; misses Absenteeism	M
Many classes with few records, or multimodal classes	Avoid NC.	Absenteeism, Glass; no controlled test	W
Symmetric classes, every pair well apart	All converge; choose on cost (NC).	Separability sweep; Soybean, Wine. Iris fails: one close pair	M
Symmetric classes, low separability	NC is competitive and far cheaper.	Separability sweep; Lenses only in real data	W
Common scale, moderate separability, many attributes	NCN is worth trying.	Ionosphere only, robust to scaling	W

use Table 5.4 to shortlist one or two methods; and confirm the choice with a corrected resampled t-test. This answers RQ3, within the limits below.

5.9 Threats to validity

Internal validity. Five of our fifteen datasets differ from the original's (Section 4.1): Thyroid is a different dataset, Absenteeism a different task, Lenses lacks one attribute, and Hepatitis and Lung cancer lack their incomplete records. Our Ecoli matches in size but not in accuracy, for reasons we could not establish. On the nine matching datasets the replication is close (Section 5.1). Original accuracies were read from a figure and are rounded, and the original's text and figure disagree on Cryotherapy.

Construct validity. The heterogeneity index uses only extreme standard deviations and ignores feature relevance, and our separability measure averages over pairs of classes, which can hide one close pair [33]. In the skewness experiment we held this ratio fixed, not the difficulty of the problem: the Bayes accuracy rose from 0.87 to 1.00, and the ratio itself drifted at the two highest levels. We therefore compare methods by their shortfall from the Bayes rate.

Statistical conclusion validity. With fifteen datasets, the Spearman correlation is significant but the Pearson and Wilcoxon results are not, and the NC skewness trend rests on one dataset. Five datasets used fewer than ten folds, and Ecoli and Absenteeism only two, so their estimates are the least precise. We report all of these.

External validity. We used one k in the main comparison, one distance, one scaler and small tabular data. Other scalers, distances and $p \gg n$ data such as SRBCT may differ, and the synthetic runs approximate NCN.

Chapter 6

Conclusion

We set out to check whether the advantage of the nearest centroid neighbourhood rule over classical KNN, reported on raw features, belongs to the algorithm or to the preprocessing. Our replication reproduced the original results closely on the nine datasets that match the original's, and we have said where our data differ. When we standardised the features and changed nothing else, most of the advantage disappeared. On Stars, Wine and Cryotherapy it fell to within 0.008 of zero. The size of the drop followed the heterogeneity of the raw feature scales, and a synthetic experiment that manipulated scale alone reproduced the effect. On Wine, the raw gap grew with k while the standardised gap did not. The one advantage that held, on Ionosphere, came from the one dataset whose features were already on a common scale. We read this as a genuine NCN advantage. It is a real one, but it rests on a single dataset.

Adding the nearest centroid classifier showed a second, independent point. A global model is sensitive to skewness in a way local models are not. With a separability ratio held fixed, NC's shortfall from the Bayes-optimal accuracy grew to about twice KNN's as skewness increased, while NCN stayed closer to the optimum than KNN. When classes are symmetric, NC matches or beats the local methods at a fraction of the cost, and when every pair of classes is also well apart, all three converge. Many small classes and multimodal classes hurt NC as well, as Absenteeism and Glass show.

None of this shows that the original study was wrong. It shows that its conclusion depends on a preprocessing choice that most practitioners would make differently. That is the general lesson of this thesis: a classifier comparison is only as general as the preprocessing and the data it was run on. Reporting the preprocessing, testing with corrected statistics and checking the result under standardisation are cheap steps, and they would have changed the conclusion here.

There are four natural next steps. The first is to replace the heterogeneity index with one that weights features by their class relevance, since the Breast Cancer

case shows that scale alone is not enough. The second is to measure separability for the closest pair of classes rather than the average, since Iris shows that the average can hide one hard pair. The third is to repeat the comparison on the exact datasets of the original study, including new-thyroid, and to extend it to other scalers, distances and $p \gg n$ data such as SRBCT. The fourth is to test the selection guide prospectively, on datasets not used to build it. Only then can its moderate and weak rows be promoted or dropped.

Bibliography

[1] Evelyn Fix and Joseph L. Hodges. Discriminatory analysis. Nonparametric discrimination: Consistency properties. Technical Report 4, USAF School of Aviation Medicine, Randolph Field, TX, 1951.

[2] Thomas M. Cover and Peter E. Hart. Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967.

[3] Bidyut B. Chaudhuri. A new definition of neighborhood of a point in multi-dimensional space. *Pattern Recognition Letters*, 17(1):11–17, 1996.

[4] José S. Sánchez, Filiberto Pla, and Francesc J. Ferri. On the use of neighbourhood-based non-parametric classifiers. *Pattern Recognition Letters*, 18(11–13):1179–1186, 1997.

[5] Tsvetelina Mladenova and Irena Valova. Comparative analysis between the traditional and nearest centroid methods in the K-nearest neighbor algorithm. In *Proc. Int. Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*, 2022.

[6] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998.

[7] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003.

[8] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

[9] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[10] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[11] Pauli Virtanen et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

[12] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research. *Journal of Machine Learning Research*, 22(164):1–20, 2021.

[13] Odd Erik Gundersen and Sigbjørn Kjensmo. State of the art: Reproducibility in artificial intelligence. In *Proc. AAAI Conference on Artificial Intelligence*, pages 1644–1651, 2018.

[14] knn_nc_research: code, data and thesis for "revisiting KNN and centroid-based classification". `https://github.com/mdakash41/thesis-comparative-analysis-knn-nc`, 2026. GitHub repository.

[15] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2nd edition, 2009.

[16] Charu C. Aggarwal, Alexander Hinneburg, and Daniel A. Keim. On the surprising behavior of distance metrics in high dimensional space. In *Proc. Int. Conf. on Database Theory (ICDT)*, volume 1973 of *LNCS*, pages 420–434, 2001.

[17] José S. Sánchez, Filiberto Pla, and Francesc J. Ferri. Improving the k-NCN classification rule through heuristic modifications. *Pattern Recognition Letters*, 19(13):1165–1170, 1998.

[18] Yoshihiro Mitani and Yoshihiko Hamamoto. A local mean-based nonparametric classifier. *Pattern Recognition Letters*, 27(10):1151–1159, 2006.

[19] Jianping Gou, Zhang Yi, Lan Du, and Taisong Xiong. A local mean-based k-nearest centroid neighbor classifier. *The Computer Journal*, 55(9):1058–1071, 2012.

[20] Robert Tibshirani, Trevor Hastie, Balasubramanian Narasimhan, and Gilbert Chu. Diagnosis of multiple cancer types by shrunken centroids of gene expression. *Proceedings of the National Academy of Sciences*, 99(10):6567–6572, 2002.

[21] Dalwinder Singh and Birmohan Singh. Investigating the impact of data normalization on classification performance. *Applied Soft Computing*, 97:105524, 2020.

[22] Md Manjurul Ahsan, M. A. Parvez Mahmud, Pritom Kumar Saha, Kishor Datta Gupta, and Zahed Siddique. Effect of data scaling methods on machine learning algorithms and model performance. *Technologies*, 9(3):52, 2021.

[23] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

[24] Dietrich Wettschereck, David W. Aha, and Takao Mohri. A review and empirical evaluation of feature weighting methods for a class of lazy learning algorithms. *Artificial Intelligence Review*, 11:273–314, 1997.

[25] Kilian Q. Weinberger and Lawrence K. Saul. Distance metric learning for large margin nearest neighbor classification. *Journal of Machine Learning Research*, 10:207–244, 2009.

[26] Sahibsingh A. Dudani. The distance-weighted k-nearest-neighbor rule. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-6(4):325–327, 1976.

[27] Salvador García, Joaquín Derrac, José Ramón Cano, and Francisco Herrera. Prototype selection for nearest neighbor classification: Taxonomy and empirical study. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(3):417–435, 2012.

[28] David J. Hand. Classifier technology and the illusion of progress. *Statistical Science*, 21(1):1–14, 2006.

[29] David H. Wolpert. The lack of a priori distinctions between learning algorithms. *Neural Computation*, 8(7):1341–1390, 1996.

[30] Manuel Fernández-Delgado, Eva Cernadas, Senén Barro, and Dinani Amorim. Do we need hundreds of classifiers to solve real world classification problems? *Journal of Machine Learning Research*, 15:3133–3181, 2014.

[31] Alessio Benavoli, Giorgio Corani, Janez Demšar, and Marco Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through Bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017.

[32] Tin Kam Ho and Mitra Basu. Complexity measures of supervised classification problems. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(3):289–300, 2002.

[33] Ana C. Lorena, Luís P. F. Garcia, Jens Lehmann, Marcilio C. P. Souto, and Tin Kam Ho. How complex is your classification problem? A survey on

measuring classification complexity. *ACM Computing Surveys*, 52(5):107:1–107:34, 2019.

[34] Dheeru Dua and Casey Graff. UCI machine learning repository. University of California, Irvine, School of Information and Computer Sciences, 2019.

[35] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.

Appendix A

Code, Data

Repository

All code, data, results and this thesis are in a public GitHub repository:

`https://github.com/mdakash41/thesis-comparative-analysis-knn-nc`

Contents

- `README.md`: what the project is, how to run it, and which script produces which table and figure.
- `main.py`: runs every experiment in order, then builds the figures.
- `src/`: shared code.
  - `datasets.py` loads and cleans the datasets.
  - `classifiers.py` implements NCN.
  - `evaluation.py` holds the repeated cross-validation and the corrected resampled t-test.
  - `measures.py` holds the structural measures.
- `experiments/`: one script per part of Chapter 5 (below).
- `data/`: the fourteen CSV files; Breast cancer is loaded from scikit-learn.
- `results/`: the CSV output of every experiment and the figures.
- `thesis/`: this thesis as a PDF.
- `archive/`: earlier exploratory work that no number in this thesis depends on.

Script to thesis map

Script	Thesis
`exp0_datasets.py`	Table 4.1
`exp1_scaling.py`	Table 5.1, Figures 5.1 and 5.2
`exp2_mechanism.py`	Figure 5.3
`exp3_structural_profile.py`	Table 4.2, Figure 5.4
`exp4_skewness_fixed_separation.py`	Table 5.2, Figure 5.5
`exp5_separability_sweep.py`	Table 5.3
`make_figures.py`	all figures, from the CSV files

To reproduce everything, install the packages in `requirements.txt` and run `python main.py` from the repository root. The exact NCN on Thyroid (7,200 records) takes the longest, roughly ten to fifteen minutes on a laptop.